COLLEGE OF ARTS AND TECHNOLOGY

CrisisNet: Disaster Response Coordination and Logistics Platform

A Real-Time System for Streamlining Disaster Relief Efforts and Resource Distribution

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TABLE OF CONTENTS

I. Introduction

1.1 Purpose

1.2 Scope

1.3 Definitions, Acronyms, and Abbreviations

II. Overall Documentation

2.1 Product Perspective

2.2 Product Functions

2.3 User Classes and Characteristics

2.4 Operating Environment

- Design and Implementation Constraints

- Assumptions and Dependencies

III. Specific Requirements

3.1 Functional Requirements

3.2 Non-Functional Requirements

3.3 Performance Requirements

3.4 Security Requirements

3.5 Usability Requirements

3.6 Reliability and Availability

3.7 Assumptions and Dependencies

IV. External Interface Requirements

4.1 User Interface

4.2 Hardware Interface

4.3 Software Interface

4.4 Communication Interface

V. System Features

5.1 Feature 1: Real-Time Resource Tracking

5.1.1 Requirements

5.1.2 Acceptance Criteria

5.1.3 Use Cases

5.2 Feature 2: Communication and Collaboration Platform

5.2.1 Requirements

5.2.2 Acceptance Criteria

5.2.3 Use Cases

5.3 Feature 3: Volunteer Management

5.3.1 Requirements

5.3.2 Acceptance Criteria

5.3.3 Use Cases

VI. Acceptance Criteria

6.1 System Testing Strategies

6.2 Metrics for Acceptance

VII. Appendices

7.1 Glossary

7.2 Related Documents

7.3 Diagrams and References

INTRODUCTION

1.1 Purpose

This software requirements specification's (SRS) goal is to give a thorough explanation of the system

proposal's functional and non-functional necessities. In times of emergency, this system is intended to improve the flow of information, coordination, and allocation of resources between volunteers, government agencies, and non-governmental organizations (NGOs) and rescue and recovery units. In order to assure successful emergency response initiatives, it seeks to centralize real-time data exchange, choice making instruments, and logistical management.

The following users are expected to utilize this paper:

- **System Engineers and Developers:** The goal is to comprehend the functional and technological conditions for system implementation.
- **Project managers and Interested Parties:** To guarantee that the system achieves the objectives of enhancing communication for disaster response.
- **System Testers:** To establish the foundation for system assessment and evaluation. ● **NGOs and Rescue and Recovery Units:** To comprehend the potential of the system and the way it will be applied in practical scenarios.

1.2 Scope

With real-time data on individuals, impacted areas, and the resources that are accessible, the CrisisNet system is intended to give a comprehensive solution for coordinating emergency response efforts. The software will streamline actions and enable smooth cooperation during emergency assistance procedures through connection with current government records and rescue service systems.

Inclusions:

- Monitoring crisis areas in real time, as well as the accessibility of resources such as food, water, medical necessities, and troops who have been dispatched.
- NGOs, contributors, and rescue and recovery units may all communicate through one unified site.
- A system for managing inquiries about resources and allocations that ensures resources are distributed according to necessities.
- Communication and supervision of volunteers, enabling entrusting tasks and monitoring of progress.
- Instruments for analysis and documentation to shed light on the efficiency of emergency aid.

Exclusions:

- The approach concentrates only on emergency response operations and fails to manage rebuilding after a disaster or recovery operations.
- With the exception of broad data on medical supply necessities, the system will not handle or retain any private or sensitive health information.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS**: Software Requirements Specification
- **NGO**: Non-Governmental Organization
- **API**: Application Programming Interface
- **SaaS**: Software as a Service
- **Real-Time Data**: Data that is provided with no interruption, right away upon compilation ● **Volunteer Management**: Managing the participation of those donating contributions to emergency relief operations
- **Resource Allocation**: Providing food, medical equipment, and other goods to areas where they are most desperately required is the procedure of resource distribution.
- **Disaster Zone**: A location afflicted by an environmental or man-made calamity such as an earthquake, flood, or wildfire.

OVERALL DOCUMENTATION

2.1 Product Perspective

The SaaS CrisisNet platform is made to work with the current emergency response facilities, such as platforms for communication, government information systems, and NGO systems. In order to provide a unified and effective calamity response, it serves as a headquarters that collects data from multiple parties, including volunteers, rescuers, and relief organizations.

In order to obtain real-time information on the accessibility of resources such as food and medical equipment, the magnitude of the crisis, and the impacted areas, the system communicates with outside data sources. For precise awareness of surroundings, it also establishes connections with weather forecasting providers and geographic mapping applications. It also connects with messaging apps including SMS, email, and notifications to help teams coordinate and communicate alerts to one another.

2.2 Product Functions

The system provides several high-level features:

- Real-time resource management keeps an eye on the accessibility of equipment and necessities like food, water, and healthcare equipment. It also provides data regarding stock placements and levels.
- Information exchanged by volunteers, agencies, and emergency response units through a shared network of communication.
- Volunteer Management: Establishes operations according to requirements and capabilities by managing volunteer assignments, places, and accessibility.
- Resource Allocation: Depending on demands from areas affected by disasters, distributes supplies immediately to the most high-priority needs.
- Creates reports in real time and after a disaster about the efficiency of rescue operations, the use of resources, and the general outcome of the response.

2.3 User Classes and Characteristics

Various user groups with different requirements are supported by the system:

- **Emergency Response Team**: They oversee the allocation of resources, oversee areas affected by disasters, and plan operations. Real-time access to information about impacted areas, volunteer rollouts, and resources at hand is needed.
- **Volunteers**: The website is used by volunteers to sign up for work, get assignments, and monitor how they are progressing. For the purpose of managing their schedules and corresponding with coordinators, they require an interface that is easy to use.

- **Government Officials**: In order to ensure that regulations are followed as well as make knowledgeable choices regarding policies, such users handle emergency preparedness on a larger scale and need access to thorough reports and analytics.
- **IT and System Administrators**: In charge of protecting the safety of sensitive information while sustaining the system's foundation. Users' controls, tracking features, and system logs must be accessible to them.

2.4 Operating Environment

Utilizing a cloud-based design, the CrisisNet platform will be able to manage immense amounts of data in calamity situations because of its high availability and scalability. Important components of the operational setting include the following:

- **Hardware**: Mobile devices, laptops, and Personal Computers are all capable of using the system. Devices that are strong and have satellite communication can be used for remote access in regions affected by disasters.
- **Software**: In order to ensure the ability to grow and adapt, the system will be developed using present web standards such as HTML5, CSS3, and JavaScript and a cloud-based backend, such as AWS or Microsoft Azure. Major smartphone operating systems and software programs such as Chrome, Firefox, and Safari will be supported by the system.
- **Dependencies**:
    - **External API**: Enabling connection to systems for immediate information, navigation equipment, and forecast services.
    - **Communication Tools**: Combination of automated notification systems for mobile devices, email, and SMS.
    - **Data Sources**: Databases maintained by the government that track resources and statistics on calamities.

SPECIFIC REQUIREMENTS

3.1 Functional Requirements

- 3.1.1 Real-Time Resource Tracking

  - **Description**: The system will allow a permanent picture of the location, quantity, and status of its resources. This includes the movement of food, water, medical supplies, and personnel.
    - **Input**: Resource data (type, location, quantity, status) that may be collected from issued user reports as well as government databases.
    - **Process**: The system has the capability of processing received and real-time resource data, incorporating resources from various applications and providing a geographical view of the information.
    - **Output**: Monitoring of resources within the organization with the help of maps and virtual dashboards showing the resource inventory at that point in time and its active cache.
    - **Acceptance Criteria**: The information provided by the system is almost real-time and relevant to resource availability for enhancing resource management (no more than 5s contrast for authenticating).

3.1.2 Communication and Collaboration Platform

  - **Description**: The system shall promote interaction and discussion amongst all relevant parties (volunteers, agencies, and emergency response units).
    - **Inpu**t: Input Measures consisting of messages and alerts that require action from individuals and/or include pictures and/or location information.
    - **Process**: Through different communication means such as chat, alerts, other media, and discussion forums, it is possible to communicate and share information amongst users safely and reliably.
    - **Output**: Effective communication and transfer of information among all stakeholders can take place, enhancing collaboration and better decision- making.
    - **Acceptance Criteria**: Messages must be sent out quickly with few if any delays; and all communication systems must work even if heavily utilized.

- 3.1.3 Volunteer Management

  - **Description**: The system shall implement a register, allocation and monitoring of

volunteers, allowing them to register, state their areas of expertise and their

availability and be allocated particular duties.

- **Input**: Information about the volunteer (skills, availability, location, contact information) that is received through the registration process.
- **Process**: The system holds and administers the volunteers' profiles, allocates volunteers' tasks according to their competencies, schedules and geographical location as well as coordinates the implementation of the tasks.
- **Output**: Seamless mapping of volunteers skills with available tasks to assist in proper manpower planning.
- **Acceptance Criteria**: Volunteers have a straightforward approach to register or modify their details as well as undertake a task, according to their skills and the context of the task.

3.2 Non-Functional Requirements

3.2.1 Performance Requirements

- The system shall be able to allow many users at the same time (more than 10,000).
- There shall be a fast response time (less than 2 seconds) for emergency operations.

3.2.2 Security Requirements

- The user information shall be secure with appropriate encryption and access control.
- The system shall not be vulnerable in case of a security compromise or data theft.

3.2.3 Usability Requirements

- The interface shall meet the needs of all user categories and be user-friendly in their operation.

● The system shall be online on many devices (desktops, laptops, tablets, smart phones).

3.2.4 Reliability and Availability

● The system shall be dependable and dependable in terms of the system's performance (not less than 99.9% system's performance).

● The system shall have disaster recovery capability.

3.3 Performance Requirements

● There must be no degradation in performance of the system even when a large number of users (not less than 10,000) are simultaneously running the program which will ensure a good response and availability even when large scale disasters strike.

● For meanwhile or even potentially critical operations such as locating resources and sending alerts, changing the location of volunteers, the system response time shall be short enough.

● The system shall operate even at critical disaster situations and handle large amounts of information and heavy traffic without failing or slowing down.

3.4 Security Requirements

● Even the user information consisting of the personal information and the other sensitive information pertaining to disasters shall be encrypted using high encryption algorithms such as AES-256 while both in use and storage in order to preserve the confidentiality of the information.

● There must be effective access control provisions whereby access to particular areas and functionalities can only be provided to specific users in accordance with their designations.

● The system will be subjected to routine penetration testing and regular security assessments so as to manage all vulnerabilities, cyberassaults and data breaches in order to protect the health and safety of users' information within the system.

● The system shall protect user information with strict measures against infringing any data privacy policies and guidelines.

3.5 Usability Requirements

● The user interface shall be standard, conventional, and the navigation process will be simple and straightforward for every classified user irrespective of his or her technical skill.

● The system shall ensure that users receive sufficient, effective informational instructions on how to undertake these processes and find such resources as is necessary to cut any confusion and errors.

● The system should be used on multi-devices (desktop, laptop, tablet, smart phone) with fluidity so as to suit all different users in different situations depending on their screen sizes.

● The system should comply with the design standards and be suitable for use by all including the physically challenged and are therefore bound by legal regulations (e.g. accessibility standards by WCAG).

3.6 Reliability and Availability

● The system shall be dependable and will provide satisfactory availability measures of a minimum of 99.9% more than before to ensure working continues even in extreme disaster when ever needed. Twin redundancy methods, and failover systems, etc should be put into place to reduce the down times and eliminate interruptions in the service.

● The system shall incorporate provision of disaster recovery measures which will allow the business processes to continue in the case of unanticipated incidents such as weather or community attacks.

● Regular backups of critical data shall be performed and stored securely to enable quick recovery in case of data loss, ensuring data integrity and minimizing service interruptions.

3.7 Assumptions and Dependencies

● Assumptions:

○ Assumptions are made that the target users will be able to get a stable internet allowing them to operate the system effectively including in instances where disaster occurs and communication is important. ○ It is also assumed that the government institutions and other bodies that provide data will ensure that their data will

make it available through resources such as APIs so that the system can be able to load information that was not previously served.

○ It is expected that registration users shall be concerned on inputting correct information

and when using the system, which maintains data validity and helps in proper resource management.

- Dependencies:
  - It is also recognized that the system requires interactive and integrated functionalities with real time weather sources, government databases, weather reporting sources and communication tools.
  - The system infrastructure availability of reliable hardware and infrastructural support such as servers and networks contributes to the system being able to work uninterrupted and avoid interruption of service delivery.
  - The system relies on the active engagement of willing participants and other stakeholders so as

to provide timely and accurate information when disaster strikes, hence contributing to the usefulness of the platform.

External Interface Requirements

4.1 User Interfaces

- **Design and Usability:** The user interface will primarily be web-based, designed to be intuitive and user-friendly. This means clear navigation, easy-to-understand icons and labels, and a responsive design that adapts to various screen sizes (desktops, laptops, tablets, smartphones). Different user roles (volunteers, emergency responders, administrators) will have customized views with functionalities relevant to their tasks.
- **Key Elements:**
  - **Maps:** The interface will prominently feature maps to display real-time resource locations and disaster zone information.
  - **Communication Tools:** Integrated communication features like in-app chat and alert systems will enable seamless collaboration.
  - **Multi-language Support:** The interface will support multiple languages to cater to users in different regions.
  - **Help and Support:** A dedicated help section will provide FAQs, tutorials, and contact information for user support.

4.2 Hardware Interfaces

The CrisisNet system will need to interact with a range of hardware:

- **User Devices:** The system must be accessible from various devices with different operating systems (Windows, macOS, Android, iOS). This includes desktops, laptops, tablets, and smartphones.
- **Specialized Equipment:**
  - **Ruggedized Devices:** For field use in disaster zones, compatibility with ruggedized devices is essential.
  - **GPS Devices:** Integration with GPS devices will allow for accurate location tracking of resources and volunteers.
  - **Printers:** To generate hard copies of reports or maps.
  - **Scanners:** For efficient data entry of paper-based documents or forms.

4.3 Software Interfaces

CrisisNet will need to integrate with other software applications:

- **Real-time Data Sources:**
  - **Weather Forecasting Services:** APIs from services like OpenWeatherMap will provide current weather conditions and forecasts.
  - **Government Databases:** Access to government databases will be crucial for obtaining population data, disaster zone mapping, and resource availability.
  - **Communication Platforms:** Integration with platforms like Twilio will enable SMS alerts and notifications.
- **Existing Systems:** The system may need to integrate with existing disaster management systems used by government agencies or NGOs for data exchange and interoperability.
- **Social Media Platforms:** Integration with social media platforms may be considered for broader information dissemination and public communication.

4.4 Communication Interfaces

- **Protocols:** The system will use secure communication protocols (e.g., HTTPS) to protect data transmitted between the user's device and the server.
- **Data Formats:** Standard data formats (e.g., JSON, XML) will be used for efficient data exchange between CrisisNet and external systems.
- **Channels:**
  - **In-app Chat:** For real-time communication between users within the CrisisNet system. ○ **Email:** For important updates and alerts.

- **SMS:** For urgent communication, especially in areas with limited internet access. ●

Multimedia Support: The system will support multimedia communication, allowing users to share images, videos, and documents.

SYSTEM FEATURES

5.1 Feature 1: Real-Time Resource Tracking

The system should track and monitor resources such as food and water, as well as medical supplies, in real time. This is to ensure that the location of these resources and their current status can be reached whenever necessary for strategic purposes by the response teams. For that purpose, integration with external data sources is a must in order to get the current state of the available resources and their location.

5.1.1 Requirements

It should contain a function that enables users to locate and verify the status of essential supplies such as food, water, and medicine in real-time. In this way, the response teams will manage to respond efficiently, as they will have the information on where these resources are and how many are available. There is a need for integration with other data for sourcing live information about the available resources, their status, and whereabouts.

5.1.2 Acceptance Criteria

This feature will achieve operational success when the system effectively tracks and displays resource availability with an update frequency of less than 5 seconds. Geographic Information Systems (GIS) for Resources must include Location, Stock levels, and Movement of resources.

5.1.3 Use Cases

Use Case 1: Remaining medical materials in the disaster zone is investigated by a logistics manager to evaluate if any more supplies are required.

Use Case 2: A response team tries to find the location of the water shipment as well as its journey to the affected location and its status.

5.2 Feature 2: Communication and Collaboration Platform

A collaboration mechanism enabling communication between volunteers, government, NGOs, and disaster response needs to be included in the system as a real-time information system. There is a need for instant messaging, file transfer, task allocation, and deliverables so as to enhance the coordination efforts in instances of emergencies. This will make it possible for the users to disseminate information, delegate work, and work together in real time and positively enhance the response to the situation.

Also, the platform ought to be complemented by external communication means like email, SMS, mobile application notifications etc so that users aren't attracted to one place and device only. At the same time, the system should also address the issue of confidentiality or protection of sensitive information within the communication by employing the use of encryption and access control.

5.2.1 Requirements

The system shall ensure appropriate interaction between all the participants including volunteers, NGOs and government institutions. It must be capable of providing these features: messaging, sharing of files, and task management in real time. The platform must also include measures of data safety and control of user access in order to avoid sabotage.

5.2.2 Acceptance Criteria

The platform will be operable when the support of a messaging feature during which the users experience no breathing space between message and reply with no more than two seconds is possible; file exchange is facilitated in an endeavor where all the files are securely hosted and task notifications addressed to all relevant users can be sent. The messages must be encoded for safety reasons.

5.2.3 Use Cases

Use Case 1: Volunteers provide status updates constantly and receive information from command headquarters regarding operations as they are assigned and information about the basics of the operation.

Use Case 2: A report on allocation of resources is provided by a government representative to an NGO in order to be able to make use of them together.

5.3 Feature 3: Volunteer Management

CrisisNet, which is a disaster response system, has inbuilt a Volunteer Management where volunteers are taken care of from the time of registration to their assigned tasks and up to their activities usages. This system caters for the proper utilization of volunteers and the management of their operations during times of disaster relief.

5.3.1 Requirements

The system shall allow compliance and deter violations of workers who perform volunteer duties. This

functionality shall provide features so that volunteers can visit the website, make their presence known, and indicate their readiness to undertake tasks. The system should have balances on the activities of the volunteers and their semester long progress, fully utilizing the tasking system.

5.3.2 Acceptance Criteria

This feature will be useful once volunteers register into the system, accept the tasks assigned to them, and submit the report at every stage of the task in real time. The system should anyway perform this feature within a very short time, seconds at best considering spatial dimensions, availability and skills.

5.3.3 Use Cases

Use Case 1: A volunteer recruits via the system as well, and further reports back via the system on completing an assigned task of distributing food aid.

Use Case 2: Communities, Non- governmental organization aid organizations, mobilizers keep track of the activities of the volunteers and change the assigned duties accordingly concerning the current state of the volunteers.

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ACCEPTANCE CRITERIA

6.1 System Testing Strategies

In order to reach the objectives in the Volunteer Management feature, the following system testing strategies will be employed to avoid any issues and to guarantee that the features are working as intended and that all acceptance criteria are met:

- **Unit Testing**: The registration process, task assignments as well as progress updates of individual volunteers will be evaluated one at a time in order to validate that it is functional and is meeting set specifications.
- **Integration Testing**: This is testing among sections to establish that they interact comfortably without challenges. This will include inter-module testing of the task assignment module and the notification system to ensure that the volunteers receive alerts sent by the task assignment module.
- **User Acceptance Testing (UAT)**: It is also an approach whereby the willing volunteers along with some of their managers will be allowed to interact with the system, as would be the case in real time settings. This should assist in identifying any aspects that need to be enhanced in the system or to reveal if there are any functions that have been overlooked and ought to be included in the system before it is made active.
- **Load Testing**: The system is then going to be put through the proclaimed upper limit of usage within which it can be expected to be only in theory as looking at how this system is going to be used logic gives no hope that it will be actually put to such use. This involves checking not only the load time, but also the functional stability of the system with a simulation of many registrants logging in, undertaking jobs, and submitting task reports all at once.

- **Security Testing**: However, further specific measures will be taken targeting distinct vulnerabilities as part of the other programmed activities apart from over guarding volunteer information from internal trespasses and intrusion.

6.2 Metrics for Acceptance

To determine how ready the Volunteer Management feature is for deployment, the following metrics will be proven and tested:

- **Registration Success Rate**: The percentage of volunteers who manage to register and create profiles without any errors should not be less than 95%.
- **Task Assignment Efficiency**: The assigned tasks must be done within the 10 seconds frame maximum, ten percent of the assignments would have been done manually showing the degree of automation.
- **Completion Rate**: The system should be able to complete at least 85% of the total number of tasks that were set forth which will show effective involvement of volunteers relative to the tasks allocated.

● **Volunteer Feedback Scores**: The feedback obtained from volunteers after performing the task should average 4 out of 5 or higher in order for the tasks to be classified as satisfaction. ● **Data Security Incidents**: In the first deployment of the system, the security breach should be zero or no incidence unauthorized system access to show the data is well protected. ● **User Engagement Metrics**: Furthermore, at least 75% of the registered volunteers would be expected to actively use the system at least once in a week (regular use) even after registering which is aimed at maximizing records on participation within the system.

APPENDICES

7.1 Glossary

- **Application Programming Interface (API)**: A collection of methods and instruments for creating software programs and incorporating outside systems.
- **Geographic Information System (GIS)**: A system that, when applied to spatial data, generates,

organizes, examines, and maps various kinds of data.

- **CrisisNet**: The suggested system name for managing volunteers, keeping records of supplies, and enabling collaboration during a crisis.

● **JavaScript Object Notation (JSON)**: A simple, lightweight style for exchanging information that is simple for computers to understand and produce as well as for people to read and write. ● **Comma-Separated Values (CSV)**: A straightforward file format for plain text storage of information in tables from sources like databases and spreadsheets.

● **Hypertext Transfer Protocol/Secure (HTTP/HTTPS)**: The information distribution procedures that enable safe encryption of information between a website and a user's web browser. ● **SSL/TLS**: Secure Sockets Layer/Transport Layer Security. Procedures designed to protect data transfer through the internet.

- **SMS**: Short Message Service. A feature in mobile devices or the internet that provides text messaging services.

7.2 Related Documents

7.2.1 Research Papers

1. ***Enhancing Disaster Preparedness, Emergency Response and Resource Allocation using Big Data Integration and Predictive Analytic***

   A research study on how real-time data could improve the efficacy of emergency

   response.

2. ***Considerations About Engaging Volunteers In Crisis Management***

   The crucial role that volunteers play and the approaches that work best for including them in crisis management.

7.2.2 Standards and Guidelines

1. ***What is the Sendai Framework for Disaster Risk Reduction?***

   An extensive handbook on global norms and procedures for calamity risk reduction and control.

2. ***FEMA National Response Framework***

   Official protocols defining the country's response to disasters and emergency situations of all kinds.

7.2.3 Technical Specifications

1. ***CrisisNet Technical Architecture Document***

   Complete information regarding the CrisisNet system's database construction, technological stack, and system framework.

2. ***API Requirements for Integrating an API***

A paper outlining the CrisisNet APIs that are used to integrate external services such weather information and communication services.

7.2.4 Previous Studies and Reports

1. ***Philippines Enhancing Disaster Preparedness, Response Capabilities***

   The article outlines the Philippines' initiatives to improve its capacity for calamity mitigation and response through cooperation with foreign organizations and innovative technologies.

2. ***DND Prefers Boosting Disaster Response Over Creating New Agency***

   The Department of National Defense has expressed its desire in the article to improve disaster response expertise rather than establish a new crisis management organization in the Philippines.

7.3 Diagrams

7.3.1 System Architecture Diagram

- Web-based front-end which can be accessed through web browsers and mobile applications.
- Back-end servers are in charge of processing and storing data.
- External systems integrations

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